EXPERIMENT 20:

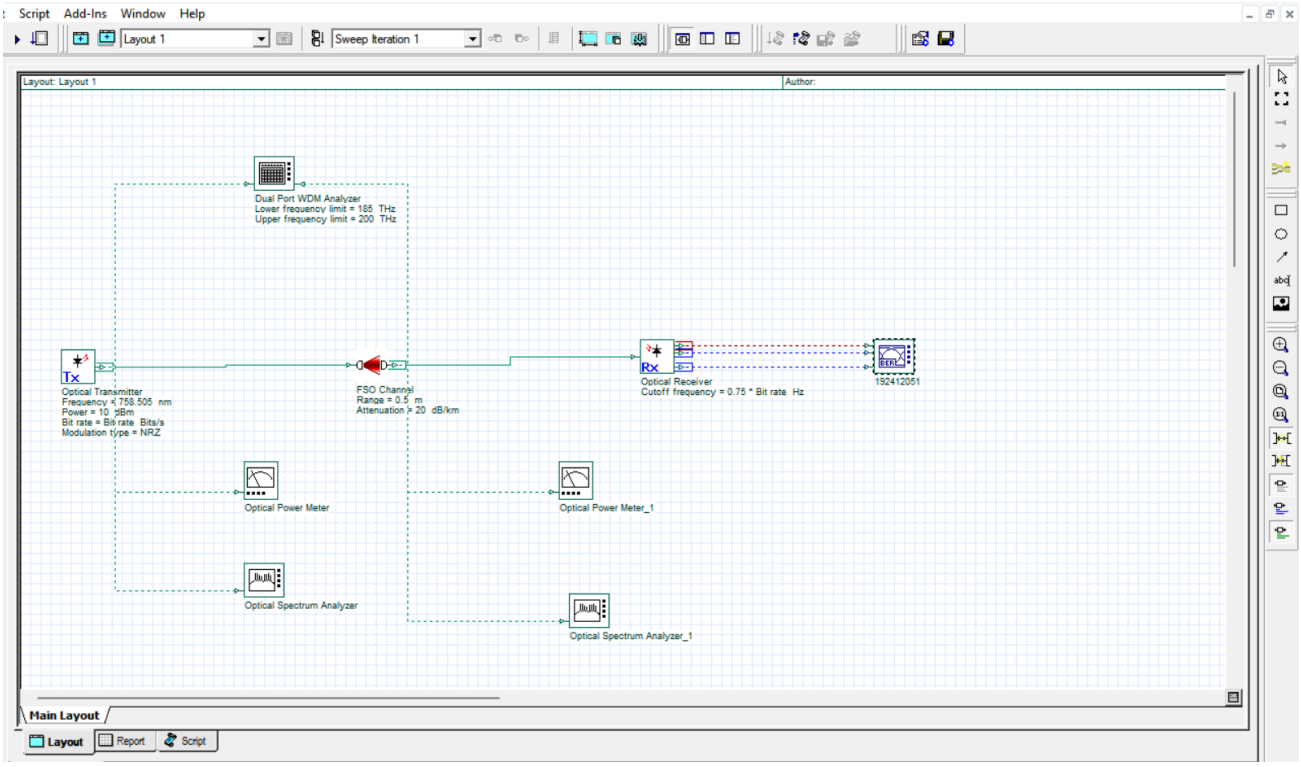
PERFORMANCE EVALUATION OF FREE-SPACE OPTICAL (FSO) LINK UNDER ATMOSPHERIC TURBULENCE USING OPTISYSTEM

OBJECTIVES :

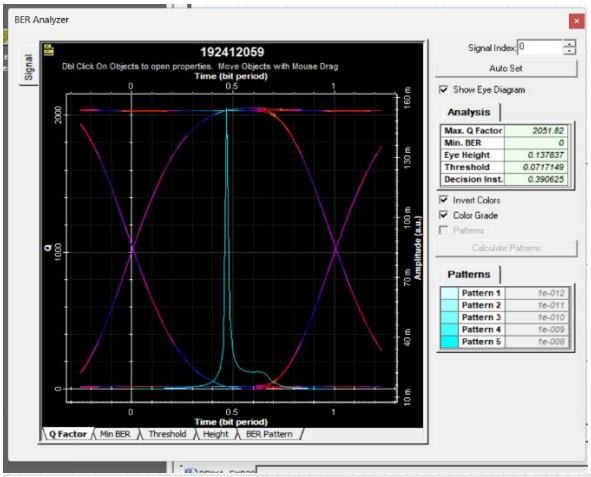
1. To design and simulate a Free-Space Optical (FSO) communication link using OptiSystem software.
2. To study the transmission of optical signals through free space.
3. To analyse the effects of atmospheric turbulence on signal quality.
4. To evaluate the performance of the FSO link using BER and received optical power.
5. To understand how atmospheric conditions affect the reliability of optical communication.

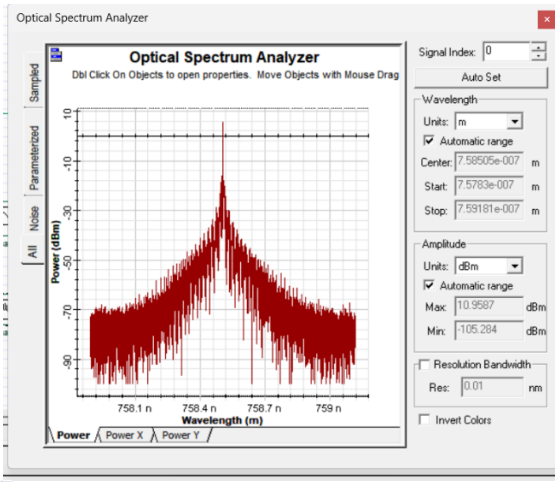
SOFTWARE REQUIRED: Opti System.

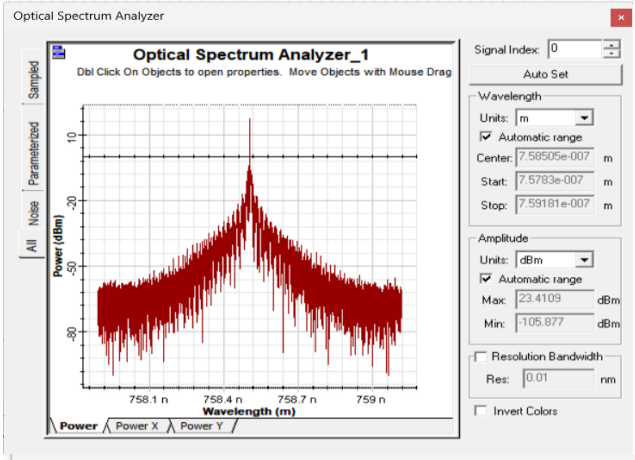
CIRCUIT DIAGRAM:



Output:


BER Analyzer
192412059
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 2051.82
Min. BER 0
Eye Height 0.137837
Threshold 0.0717749
Decision Inst. 0.390625
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


Optical Spectrum Analyzer
Optical Spectrum Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Power (dBm)
Wavelength (m)
Signal Index: 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 7.58505e-007 m
Start: 7.5783e-007 m
Stop: 7.59181e-007 m
Amplitude
Units: dBm
Automatic range
Max: 10.9587 dBm
Min: -105.284 dBm
Resolution Bandwidth
Res: 0.01 nm
Invert Colors
Power
Power X
Power Y


Optical Spectrum Analyzer
Optical Spectrum Analyzer_1
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Power (dBm)
Wavelength (m)
Signal Index: 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 7.58505e-007 m
Start: 7.5783e-007 m
Stop: 7.59181e-007 m
Amplitude
Units: dBm
Automatic range
Max: 23.4109 dBm
Min: -105.877 dBm
Resolution Bandwidth
Res: 0.01 nm
Invert Colors
Power
Power X
Power Y


Optical Power Meter
5.430 E-3 W
1.348 dBm
Signal Index: 0
Total Power


Optical Power Meter
83.310 E-3 W
19.207 dBm
Signal Index: 0
Total Power

REFLECTION:

- Learned the basic working principle of Free-Space Optical communication.
- Understood the effect of atmospheric turbulence on optical signal transmission.
- Gained practical knowledge of designing and simulating an FSO link using OptiSystem.
- Learned to analyse system performance using BER and received power measurements.

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & FSO Link Setup	Correct selection and connection of OptiSystem components for Free Space Optics link (Optical Transmitter, FSO Channel, PIN Photodetector, Low Pass Filter, Optical Power Meter, Time Domain Visualizer, BER Analyzer).	All optical transmitter, free space channel, receiver, and visualizer components are correctly selected, connected in sequence, and configured without errors.	All components are present with minor connection or layout errors, but the FSO link remains functional.	Component selection is partially complete; minor connection or configuration errors impede optical signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of physical input parameters: Wavelength (1550 nm), Modulation (OOK/NRZ), Transmit Power, FSO Range (1 to 15 km), and Atmospheric Attenuation (2 to 10 dB/km).	All input parameters and test cases across varying ranges and attenuation values are accurately set and executed.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but multiple range or attenuation test cases are missing.	Incorrect input parameter settings or failure to run the required atmospheric test cases.	20%
Measurement & Power Monitoring	Accurate extraction and recording of received optical power, Q-Factor, BER, Eye Height, and Threshold across all range settings.	Precise data extraction across all FSO range cases, neatly tabulated with complete optical power readings and clear time-domain waveforms.	Data is collected with minor omissions in recording power meter readings, BER, or eye parameters.	Partial data collection with incomplete tables or missing optical power visualizer readings.	Inaccurate, incomplete, or missing data and measurements.	20%
Atmospheric & Link Loss Analysis	Detailed evaluation of geometric spreading loss, atmospheric attenuation, distance-dependent signal degradation, and power compensation feasibility.	Comprehensive analysis correctly relating range and attenuation variations to geometric loss, received power drop, eye height reduction, and bit error rate performance.	Clear analysis identifying basic trends in received power and eye height relative to FSO distance.	Basic attempt at result analysis without explaining physical mechanisms like atmospheric scattering or geometric loss.	Erroneous or missing analysis of simulation results and power meter outputs.	20%
Report & Conclusion	Identification of maximum reliable communication distance, power compensation trade-offs with diminishing returns, and structural clarity.	Report features clear time-domain waveforms, complete power tables, and insightful conclusions identifying optimal operational parameters under atmospheric impairments.	Satisfactory report with relevant visualizer displays and sound concluding summary.	Incomplete report with missing graphs, power readings, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

CONCLUSION :

Thus, the performance of a Free-Space Optical (FSO) communication link under atmospheric turbulence was successfully evaluated using OptiSystem. The simulation helped to understand how atmospheric turbulence affects optical signal quality and transmission. The performance was analysed using received optical power and Bit Error Rate (BER). Hence, this experiment provided a better understanding of FSO communication and its reliability under different atmospheric conditions